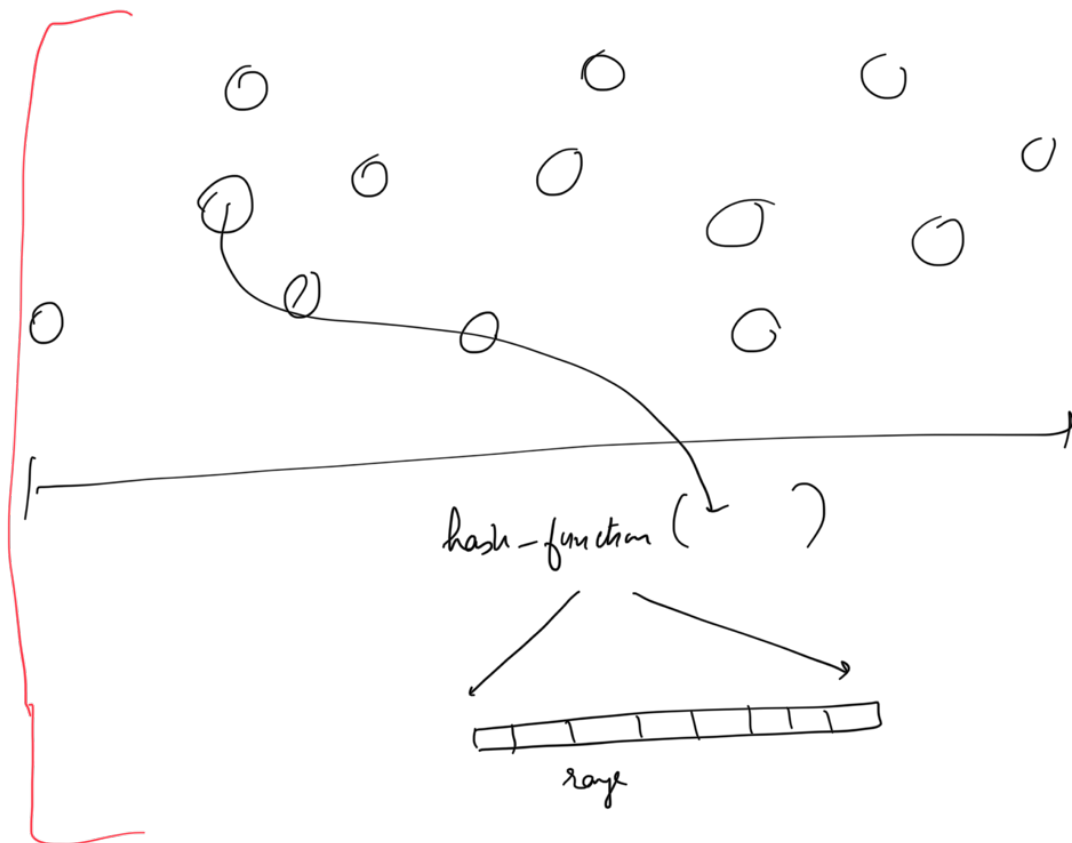


datatype Hashing $\equiv$ Concept

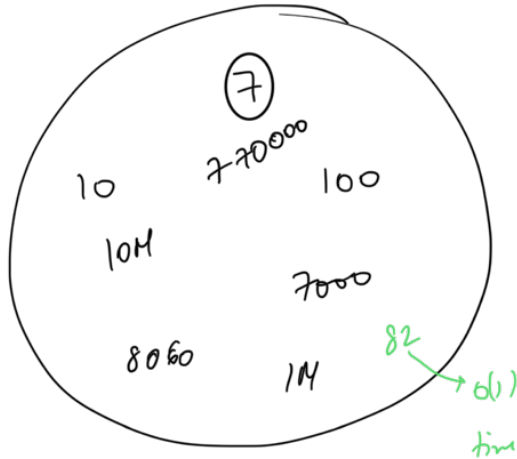
HashMap
dictionary
Set

insert a new elem
O(1) time

Find Search elem
in O(1) time



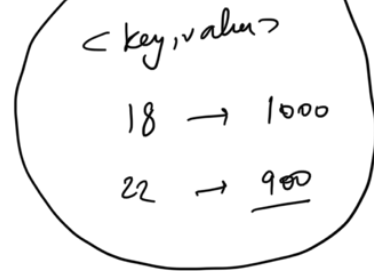
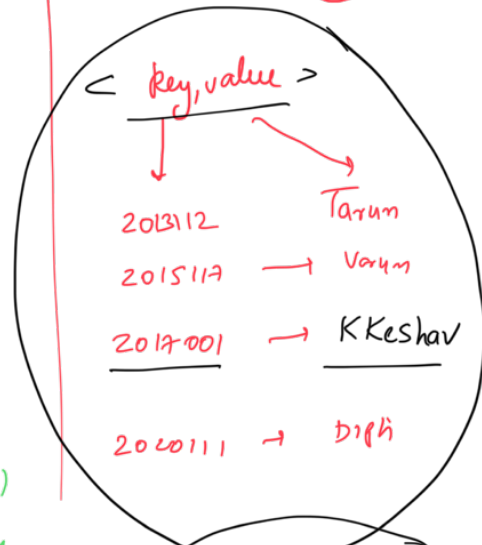
Hashset



Find(7000) → Yes d() time

Find(2000) → No d() time

HashMap



③ No order of data points

④ Uniqueness

Ⓐ add a k, v
m o (1)

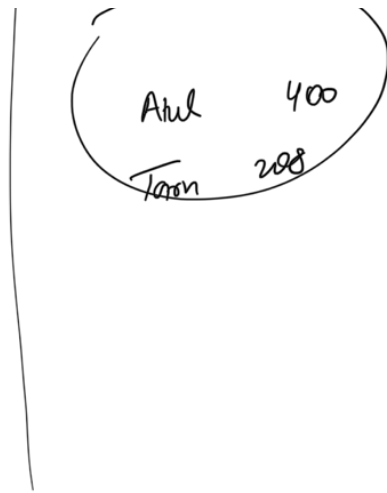
Ⓑ find m keys
in o(1) time

Ⓒ Can't search
for value

→ No order

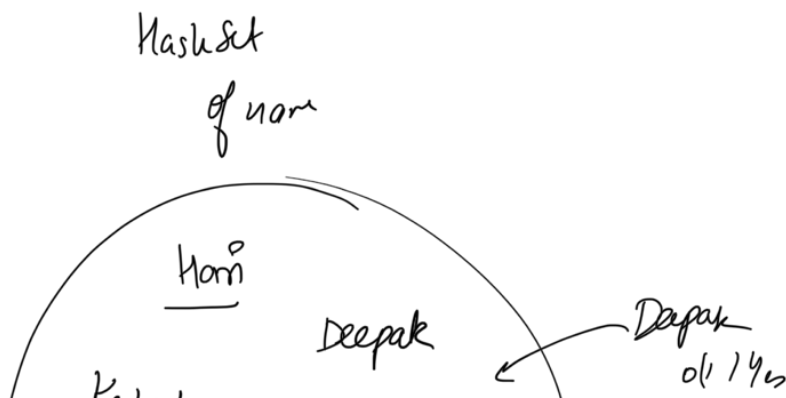
Uniqueness is m keys
by overloading

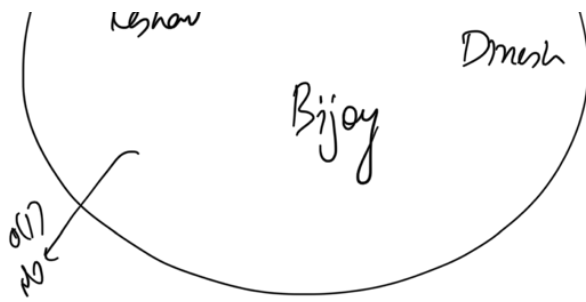
k, v



- ① Backbone concepts hashing
- ② different names
- hashmap
 - hashset
 - unordered-map
 - unordered-set
 - dictionary
 - set
 - map

③ NO ORDERING





HashMap

< key, value >

key $\equiv$ country Name

value $\equiv$ GDP

< india, y >

Q1 Given an array A

How many distinct elements exist

A: [7, 3, 2, 1, 3, 7, 0]

~~(S)~~

```
Set = {}  
for (i=0, i<N, i++)  
    Set.add(arr[i])  
print (set.size())
```

$O(1)$ ←

$T.C = O(N)$

aux S.C = $O(N)$

Q'

How many elements in your Array
are uniquely exist

A: [7, 3, 2, 1, 3, 7, 0, 3]

X X X X ↓
 ↓ ↓ ↓
 ↓ ↓ ↓

~~3~~ 3

HashSet →

HashMap → H.W

7 | 3 | 2 | 1 | (3) | 7 | 0 | 3
↑ ↑ ↑ ↑ (7)

Set
qualifying-set
7 | 3 | 2 | 1 | 0

Set
disqualify set
3 | 7

★ qs.sim() - ds.sim()

set.insert()
set.find()
set.remove(elem) } O(1)

①

queries

low can cycle shj

$$|S| \leq M$$

$q \equiv$

"abc"

Sum of ascii
values of
shj dooch



294 ~~A~~

q: "a" $\longrightarrow$ 97

q: (bcd) $\longrightarrow$ 297

$O(M)$ $\left[\begin{array}{l} \text{ans} = 0 \\ \text{for } (i = 0; i \leq M; i++) \\ \text{ans} = \text{ans} + q[i] \end{array} \right.$

$$O(QM)$$

$$\begin{array}{l} \textcircled{a} \rightarrow 97 \\ \textcircled{b} \rightarrow 98 \end{array}$$

$$\textcircled{2} \rightarrow 1122$$

$$\underline{\underline{"bca"}} \rightarrow \underline{\underline{297}}$$

$Q \gg \gg$ Possibilities



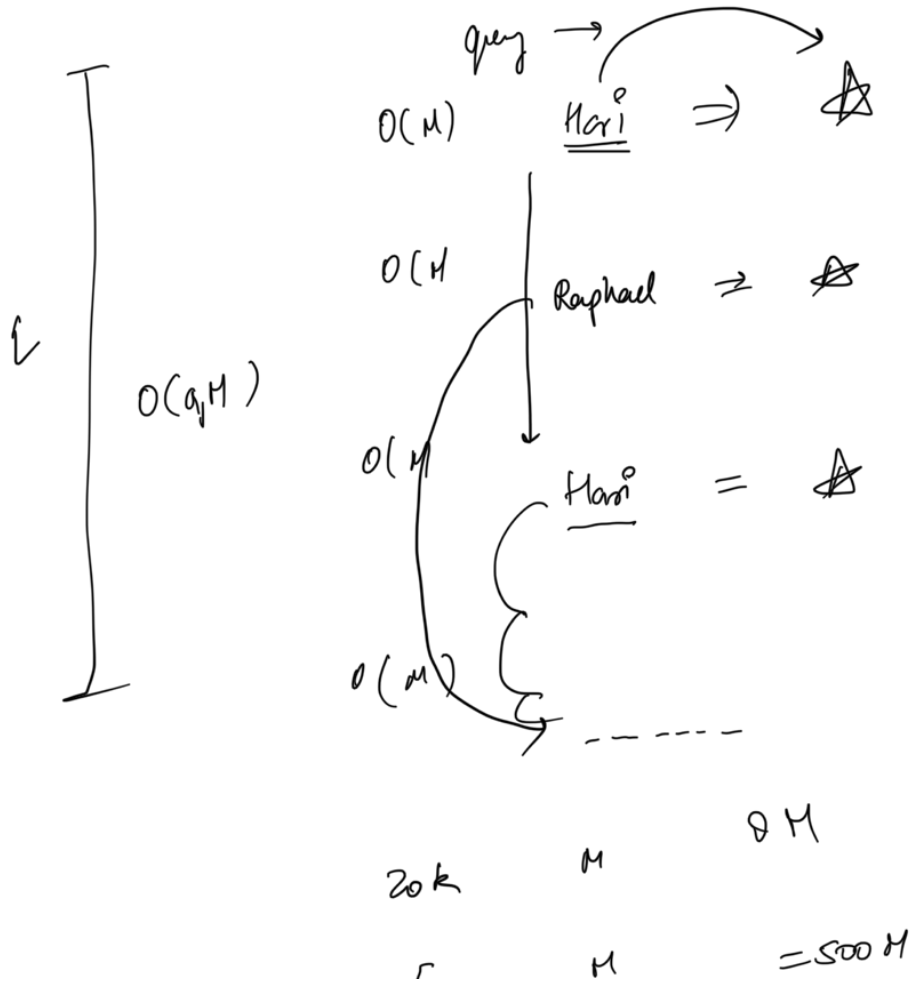
Cadix



⑧

Amazn

SDE 2



200

Unique elem $\rightarrow$ = UM

Unique Duplicates $\ll$ $\ll$ Queue
'u' 'Q'

$U < < Q$

0
Answer)

A
(A) = N

First non repeating elem of the array

A: 8 | 2 | 8 | 3 | 1 | 2 | 6 | 5 | 8

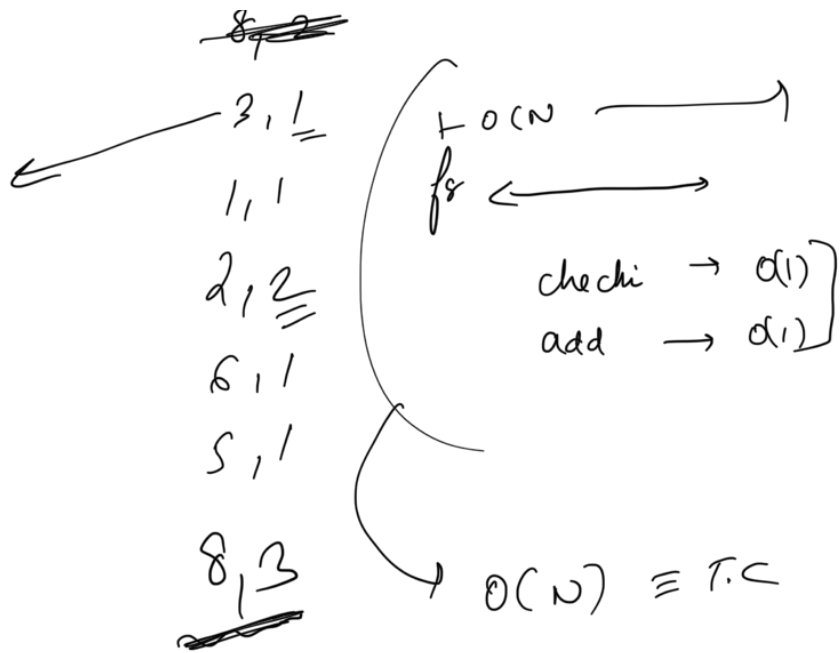
Unique



< Not, we can >

~~8~~

~~2~~



$O(N)$

$$T.C = O(N) + O(N)$$

$$= O(N)$$

aux S.C = $O(N)$

⑧

Amam /

Google /

1. .. /

Array of size N

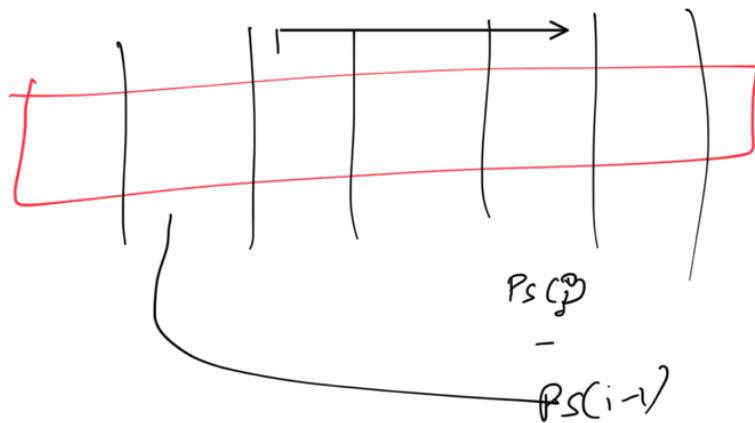
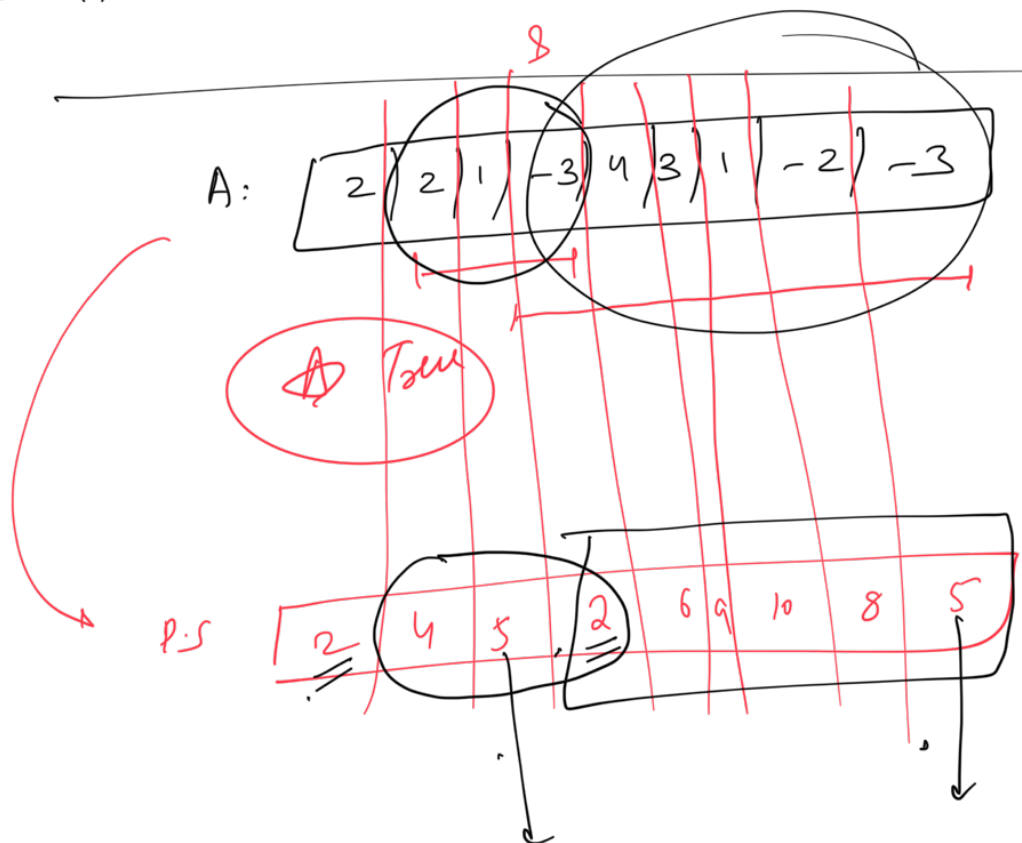
If there exists a subarray with

Microsoft /

U

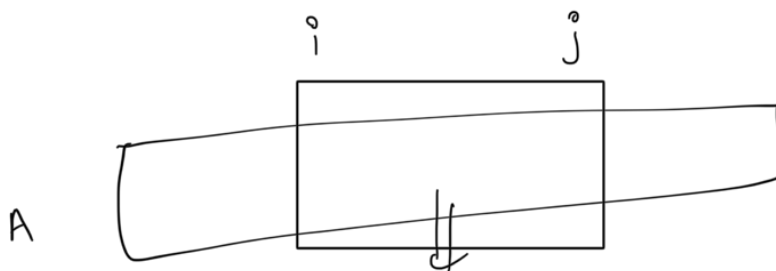
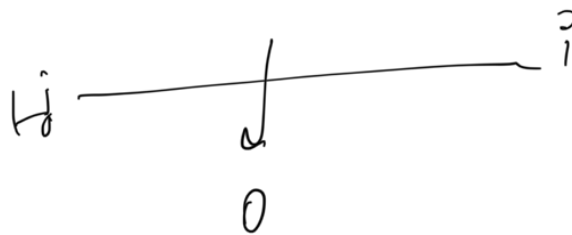
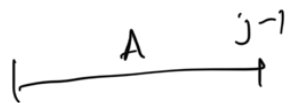
Sum = 0

LinkedIn /





$$\frac{PS(i)}{A} - \frac{PS(j-1)}{A} = 0$$

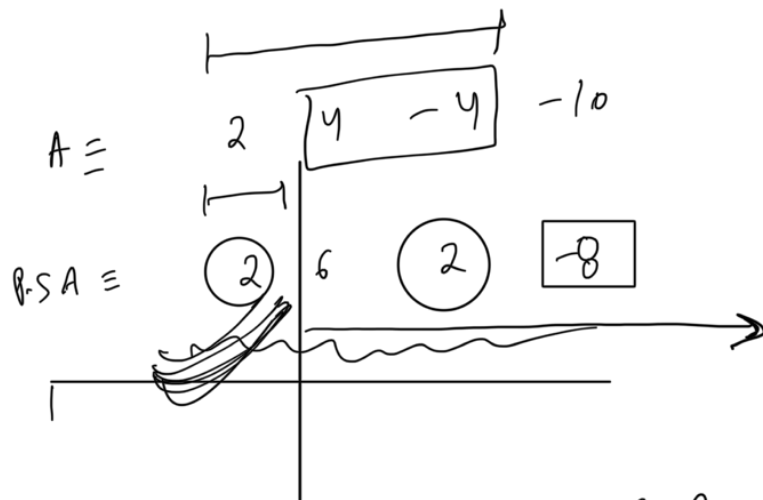


Sum = 0

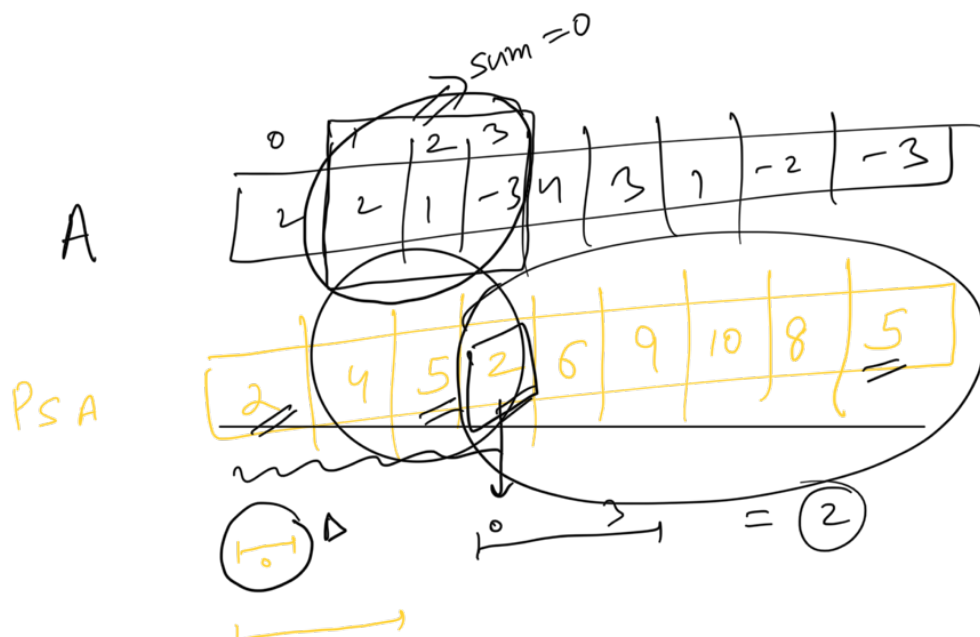
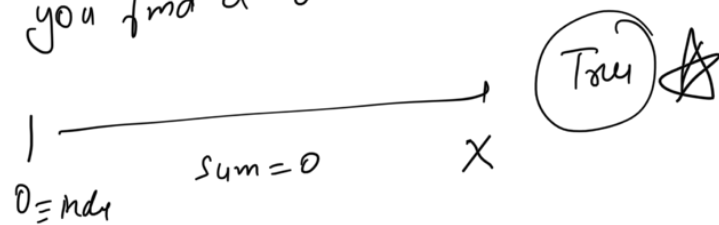
$$A \equiv 2, \boxed{2, 1, -3}, 4, 3, 1, -2, -3$$

⁰ ¹ ² ³ ⁴ ⁵ ⁶ ⁷ ⁸

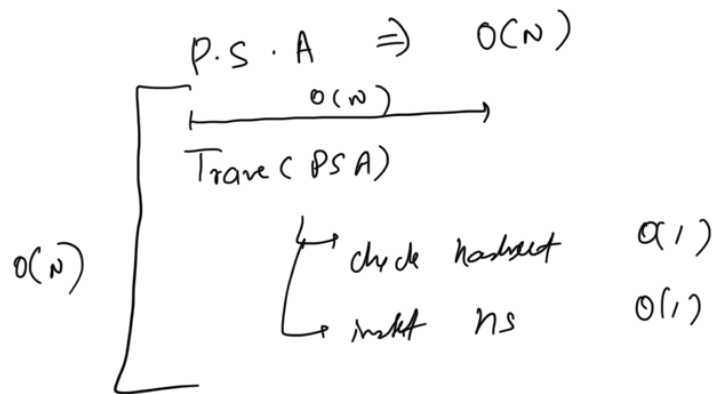
$$P.S.A \equiv 2 \quad 4 \quad 5 \quad 2 \quad 6 \quad 9 \quad 10 \quad 8 \quad 5$$



if you find a 0 in ~~set~~ P.S.A



0 1



$$T.C = O(N) + O(N) = O(N)$$

Ans S.C = $O(N)$ $+$ $O(N)$

PSA HSet

$$= O(N)$$

⑨ Facebook
Google

Two Sum

A array
value k

Check if there exists
two indices i, j

$$i \neq j$$

$$A(i) + A(j) = k$$

A 2 | 7 | 11 | 15 | 7

K=18
True

K=14
True

K=4
No

K=22
True

K=20
No

for(i=0, i<N, i++)

$$\{$$

$$\{ \text{if } (j = i+1, j < N, j++)$$

$$\{ \text{if } (A[i] + A[j] == k,$$

$$\text{return true}$$

$$\}$$

$$\}$$

$O(N^2)$

return false

hs = {}

$O(N)$

$$\left[\begin{array}{l} \text{for } (i=0; i < N, i++) \\ \text{hs.add}(A[i]) \end{array} \right.$$

logical
 read

$$\left[\begin{array}{l} \text{for } (i=0; i < N, i++) \\ \{ \text{// for } \underline{A[i]} \\ \text{to contribute to the answer} \\ \text{we need to find} \end{array} \right.$$

K

$K - A(i)$'s presence

if (h.s. has $K - A(i)$)

return True;

return False;

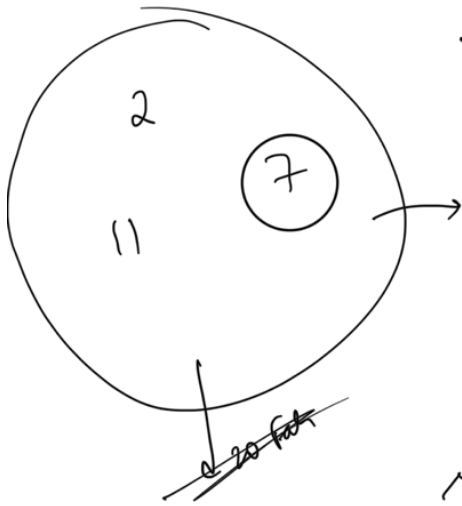
↓	2	6	8	10	2

$2 = 4$
~~True~~ False

key, value

< num, freq >

if $(A(i) == (K - A(i)))$
 if $(freq[A(i)] > 1)$



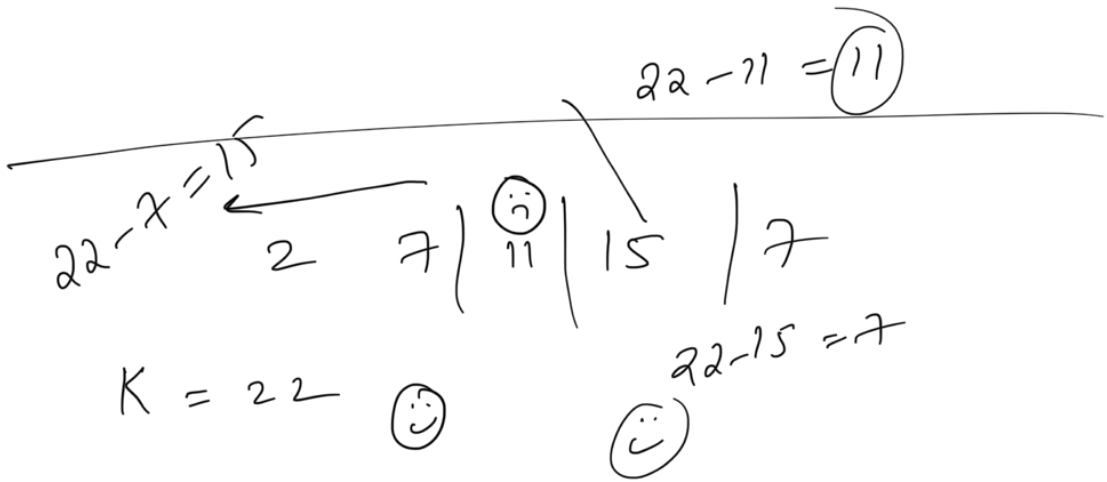
hs = {}

for (i = 0; i < N; i++)

{ if (hs.has(K - A(i)))
 return False

hs.add(A(i));

}
return True;



14

2 | 7 | 7 | 11

